

Conditional flow matching for genomic panel completion: inferring IMPACT505 alterations from the IMPACT341 targeted panel in non-small cell lung cancer

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1 Introduction

Comprehensive genomic profiling (CGP) underpins treatment selection in NSCLC, yet panel content varies widely across institutions and health-care systems. Smaller targeted panels reduce cost and turnaround time but leave part of the tumour genome unmeasured.

Objective. To determine whether a conditional generative model can infer alterations in genes absent from a targeted panel, and to quantify which biological signals make this possible.

2 Study design



Figure 1. Task definition. IMPACT341 $\subset$ IMPACT505; the 164 genes present only in IMPACT505 are masked from the input and predicted.

Table 1. Cohort and target characteristics (MSK-CHORD).

Patients, NSCLC, sequenced on IMPACT505	2,226
Input genes (IMPACT341)	341
Target genes (IMPACT505 \ IMPACT341)	164
Altered target gene–patient pairs	1.21%
Patients with ≥ 1 altered target gene	[]%
Split (patient level): train / val / test	70 / 15 / 15% [confirm]

Alteration encoding. One binary event per gene and patient: non-synonymous mutation, amplification or deep deletion. One sample per patient; [state handling of VUS and structural variants].

3 Methods

3.1 Conditional flow matching. Let $c \in \{0,1\}^p$ denote a patient's IMPACT341 profile and $x_1 \in \{-1,+1\}^{164}$ the target genes. A velocity field v_θ is trained along the linear path between Gaussian noise and data:

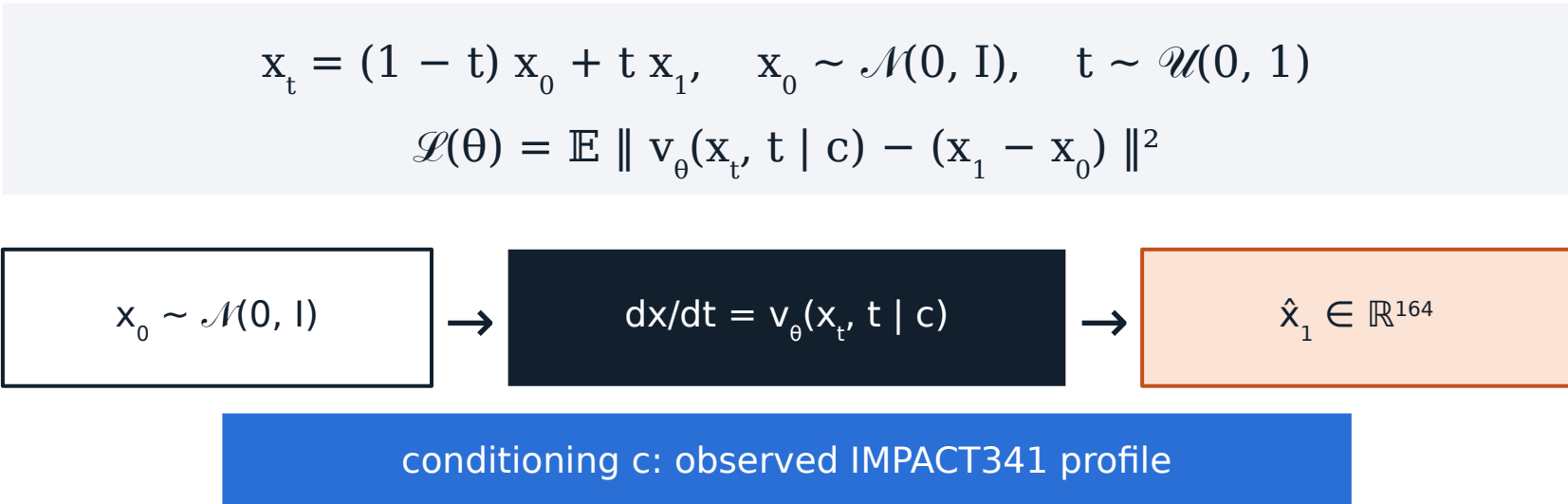


Figure 2. Inference: the learned ODE transports noise to a completed 164-gene profile conditioned on the observed panel.

3.2 Probability read-out. At $t = 0$ the path is independent of x_1 , so the optimal field satisfies $v^*(x_0, 0 | c) = \mathbb{E}[x_1 | c] - x_0$. Marginal alteration probabilities follow from a single evaluation, $\hat{p} = \frac{1}{2}(1 + v_\theta(x_0, 0 | c))$, avoiding Monte Carlo ties at $\sim 1\%$ prevalence.

3.3 Statistical analysis. Primary endpoint: AUROC on held-out patients, pooled over gene–patient pairs. Secondary: per-gene AUROC (mean over genes with ≥ 10 events), for which a prevalence-only predictor scores exactly 0.50. Comparators on identical splits: prevalence only; burden + locus (log mutation/CNA counts and copy-number state of the nearest assayed gene); L2-regularised logistic regression. 95% CIs by patient-level bootstrap (200 resamples).

3.4 External replicate. MSK-IMPACT 2017 (cBioPortal, ODbL): IMPACT341 $\rightarrow$ 69 IMPACT410-only genes in 1,215 NSCLC patients, 3 $\times$ 5-fold patient-level cross-validation with fold-stratified AUROC.

4 Results

4.1 Primary cohort. Across network architectures and training paradigms, the flow model reached a pooled AUROC of 0.77–0.79 on held-out patients, exceeding the prevalence-only baseline in all variants (Table 2).

Table 2. Discrimination on the MSK-CHORD test set. AUROC (95% CI).

Model	Pooled	Per gene
Prevalence only	[]	0.50
Burden + locus	[]	[]
L2 logistic regression	[]	[]
Flow matching	0.77–0.79	[]

4.2 Benchmark on public data. In the replicate, gene frequency alone yields a pooled AUROC of 0.715; per-gene AUROC isolates patient-specific information (Figure 3).

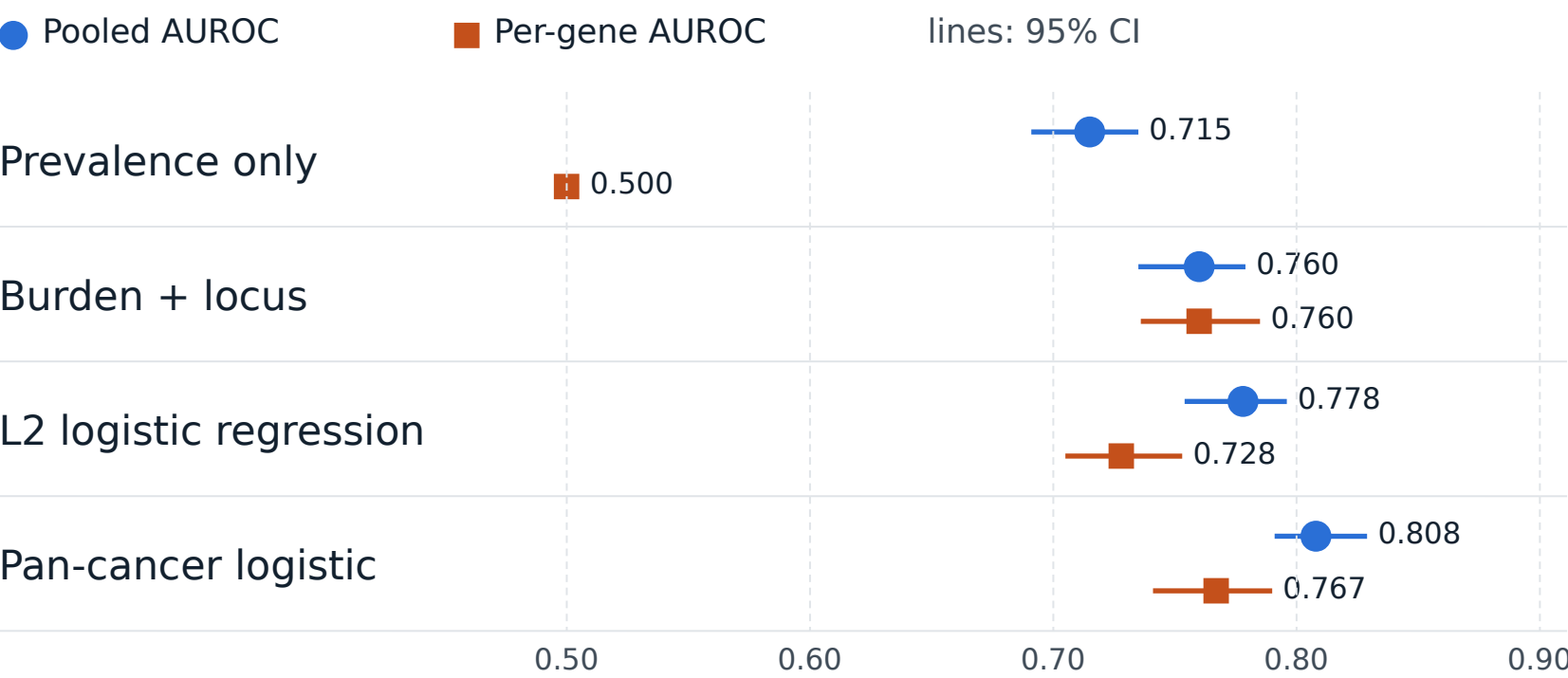


Figure 3. Replicate ($n = 1,215$): AUROC with patient-bootstrap 95% CI. Pan-cancer: trained on 7,773 patients with cancer type as covariate.

4.3 Sources of signal. Amplifications were recovered almost completely through co-amplification with neighbouring assayed genes (e.g. NFKBIA with NKX2-1 in 39/39 cases); mutation prediction matched a model using only three burden counts (Figure 4).

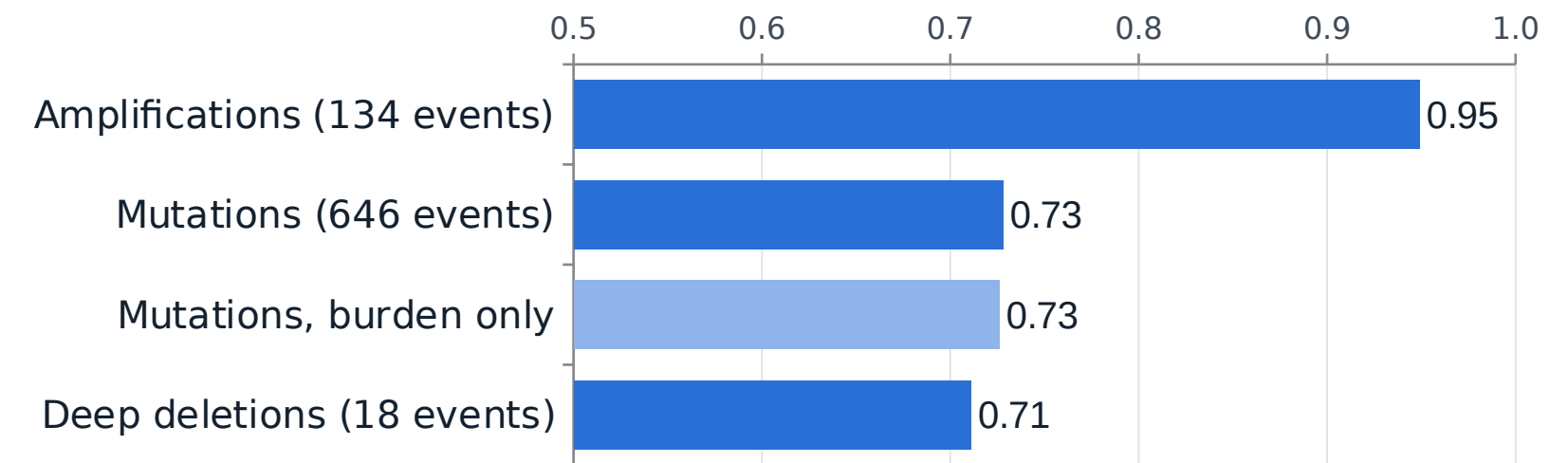


Figure 4. Replicate: per-gene AUROC by event type (positives-weighted mean; axis origin 0.50), pan-cancer logistic regression.

4.4 Simulated versus real input. In 167 patients sequenced on both panels, input from the separate IMPACT341 assay reduced discrimination relative to the masked large panel (Figure 5).

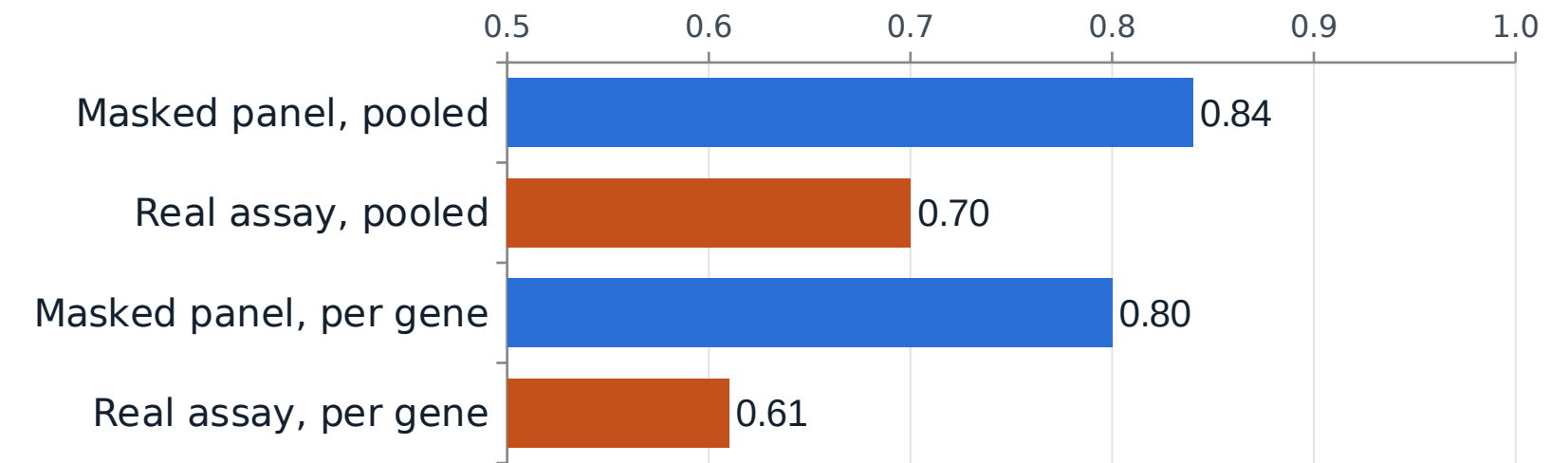


Figure 5. Replicate: AUROC with simulated (masked IMPACT410) versus real (IMPACT341 assay) input; axis origin 0.50.

5 Discussion

- A targeted panel contains recoverable information on genes outside its assayed set (pooled AUROC 0.77–0.79).
- This information derives mainly from copy-number co-location and mutational burden.
- Limitations: simulated masking overestimates performance on real assays; single test split; actionable-gene subset not yet evaluated.
- Such models may prioritise patients for broad CGP; they do not replace it.

References

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- Lipman Y, et al. Flow matching for generative modeling. ICLR 2023.
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Data and code

MSK-CHORD and msk_impact_2017 via cBioPortal. Code: [repository link]

Disclosures · Contact

[Conflict-of-interest statement]
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